

Ph.D. Applied Qualifying Exam 2018 (Take Home)

August 23 (9 AM) to August 24 (5 PM)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have technical questions about the two exam problems, you may email Prof. Sijian Wang at sijian.wang@stat.rutgers.edu (for problem 1) and Prof. Ziqiang Tan at ztan@stat.rutgers.edu (for problem 2), cc'ing Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu). If you have questions regarding logistics (e.g., submission instructions given below), please email Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

1. Type up your answers to the two problems in two separate Word or Latex files.
2. Create a folder for each problem. The folder must contain the following:
 - (a) A word or pdf file containing your answers to the problem, including all tables, figures and Appendices.
 - (b) An R (or python) file containing the codes used to generate your results.
 - (c) A word or a text file that gives a summary of the codes submitted.
3. Place the two folders into a single folder titled **yourlastname_yourfirstname_Appliedexam2018**. Zip this folder and email it to hcrane@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM August 24, Friday.

Problem 1

Background

Bone may be more sensitive to loading (applied force) during growth than in adulthood, yielding greater benefits in bone mass, geometry and strength for future fracture prevention. However, it is unknown whether loading during specific key maturity phases yields superior long-term benefits. This information is critical to the development of targeted physical activity programming for reduction of long-term osteoporosis fracture risk. The study described in the following section was conducted with the aim of obtaining such information.

Research Design and available data

This study includes 46 female subjects, who form a subset from an ongoing longitudinal study (1997-present). These subjects were recruited from local gymnastics clubs, local grade schools and girl scout troops and were divided into the following categories:

1. NON (indicated as group 2 in column F of the data file): Subjects who never had gymnastics.
2. GYM (indicated as group 1 in column F of the data file): Subjects who practiced gymnastics.

At the time of enrollment, the subjects in the NON group were matched to GYM by group for chronological age at menarche, size, and physical maturity. Each subject underwent annual whole body and regional DXA scans on dates indicated in column B of the data file. The response recorded in column M is Sub-head bone mineral content or Sub-head BMC. Several covariates were recorded for each subject at the time of each DXA scan. A detailed description of the data in the columns of the Excel file “GYM_VS_NON” is given in Section “Data description”.

Questions

1. Is this study a randomized experiment or an observational study? What is the treatment whose effect is being studied?
2. The study claims to have matched subjects in the NON group and GYM group with respect to chronological age at menarche, size and physical maturity (as measured by standing height, FA length and Sub-Head LM). Analyze the data and examine to what extent these covariates

are balanced in the NON and the GYM group. Provide a written summary of your analysis, using tables and/or graphs as appropriate and draw conclusions (**Your answer to this part should not exceed two pages**).

3. The eventual goal of this study is to investigate the association between gymnastics and bone development. Perform an exploratory analysis (that is simple summaries and plotting of the data - no modeling) as the first step to achieving this goal. Provide interpretations of the graphs and tables you present (**You answer to this part should not exceed two pages**).

NOTE: Your combined answers to parts 4 and 5 should not exceed six pages including all tables, graphs, their interpretations, comments and remarks.

4. Develop a model for investigating the association between gymnastics and bone development, explaining the logic behind your choice. Provide interpretations of the model parameters. *Note that this model development process may be iterative, in the sense that you start with a simple preliminary model, drawing inputs from the exploratory analysis, and then improve it based on model diagnostics and/or prediction performance.*
5. Provide a clearly written description of your assessment of the effect of gymnastics on bone development, using data summaries, tables and graphs as appropriate.

Data description

- ID: Subject ID number
- Group Division (GYM=1, NON=2): NON group: never exposed to gymnastics; GYM: were exposed to gymnastics.
- Covariates:
 - DXA Date: The date on which the subject got DXA scans.
 - Chronological Age at Menarche: Age of the subject when she had the menarche.
 - Menarcheal Age @ DXA: The difference between Chronological Age at DXA and Chronological Age at Menarche.
 - Date Quit Gymnastics: The date on which the subject quit gymnastics.
 - Menarcheal Age @ Quit Date: Menarcheal Age when the subject quit gymnastics.
 - Standing Height: Height of the subject.
 - FA length (cm): Length of forearm.

- Sub-Head LM (kg): Lean mass of sub-head.
 - Non-Gym Physical Activity (hrs/wk): Hours per week of the non-gymnastic physical activities.
 - Gymnastics (hrs/wk): Hours per week of the gymnastic activities.
- Sub-head BMC (Response): Sub-head bone mineral content.

Problem 2

Suppose that we observe $Y = (y_1, y_2, y_3, y_4)$ from a multinomial distribution:

$$Y \sim \text{Multinomial} \left(n; \frac{1+\theta}{3}, \frac{1-\theta}{3}, \frac{1-\theta}{3}, \frac{\theta}{3} \right),$$

where $n = 200$. Consider the data augmentation $Z = (u_1, u_2, y_2, y_3, y_4)$ such that $y_1 = u_1 + u_2$ and

$$Z \sim \text{Multinomial} \left(n; \frac{1}{3}, \frac{\theta}{3}, \frac{1-\theta}{3}, \frac{1-\theta}{3}, \frac{\theta}{3} \right).$$

(i) Derive an Expectation-Maximization (EM) algorithm based on the preceding data augmentation, to find the maximum likelihood estimator (MLE) of θ (denoted as $\hat{\theta}$).

(ii) Denote by $p(y; \theta)$ the probability mass function of Y , and by $f(z; \theta)$ the probability mass function of Z . Show that

$$\frac{d^2}{d\theta^2} \log p(Y; \theta) = E_{\theta} \left\{ \frac{d^2}{d\theta^2} \log p(Z; \theta) \right\} + \text{var}_{\theta} \left\{ \frac{d}{d\theta} \log p(Z; \theta) \right\},$$

where $E_{\theta}()$ and $\text{var}_{\theta}()$ are evaluated under the conditional distribution of Z given Y , derived from the probability mass function $f(z; \theta)$ for Z . [Note that Y is a function of Z .]

(iii) Use the result from (ii) to show that the observed Fisher information and hence an estimate of the asymptotic variance (denoted as $\hat{V}$) for the MLE $\hat{\theta}$, can also be obtained from the output of the EM algorithm in (i).

(iv) Implement the EM algorithm to compute the MLE $\hat{\theta}$ and its estimated asymptotic variance $\hat{V}$ using R. Report your results (in the form of estimate $\pm$ standard error) when it is observed that $Y = (108, 31, 27, 34)$.

(v) Design and perform a simulation study (with a fixed sample size n) to numerically investigate whether (1) the MLE of θ is approximately normally distributed, centered about the true value of θ , and (2) the variance of the MLE $\hat{\theta}$ is closely estimated by $\hat{V}$. Explain and interpret the simulation results clearly.